

Sumukh Varma Y

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SUMMARY

Backend-focused Software Engineer with 3 years of experience building and scaling distributed systems in cloud-native environments. Proven track record in architecting high-reliability pipelines, optimizing database performance, and owning end-to-end delivery across complex systems. Specializes in Node.js, TypeScript, Python, and AWS. I am known for a builder mindset; from shipping production microservices and agentic AI workflows to scaling a personal side project to 400K+ unique users and for technical ownership, systematic incident resolution, and measurable reliability improvements.

EDUCATION

Ramaiah Institute of Technology: CGPA 8.47
Bachelor of Engineering

Bangalore, India
2019 – 2023

- **Publication:** Modelling and Control of Tendon Driven Continuum Robot — IJNRD, Vol. 8, Issue 8, Aug 2023 (Impact Factor: 8.76)

SKILLS SUMMARY

- **Languages & Frameworks:** Node.js, TypeScript, Python, ReactJs, FastAPI, SQL, NoSQL
- **Cloud & Infra:** AWS (SQS, SNS, Lambda, SES, S3, API Gateway, RDS, EKS, ECR), Kubernetes, Docker
- **Databases & Tools:** PostgreSQL, MongoDB, DynamoDB, Redis, TypeORM, Git, Keycloak, Grafana

EXPERIENCE

Software Engineer — IHX (A Perfios Company)
Full-Stack Backend Engineering

Bangalore, India
Sep 2024 – Present

Orchestration Engine *Claim pipeline orchestration & rate limiting*

- Systems process 20,000 insurance claims/day with 99.99% reliability (~2 failures/day).
- Architected dynamic claim pipelines with runtime adaptation, context-based fallback routing, and Redis-based sliding-window rate limiting to prevent overload during peak traffic.
- Built pipeline timing controls enabling safe production validation on small traffic subsets before full rollout.
- Introduced DLQs and retry logic across claim creation and update flows — eliminated queue blockages where a single failed claim previously stalled 200–300 others.
- Migrated to entire application to object oriented programming structure from the pre-exisitng procedural programming structure.

Persistence Engine *Bulk writes & query optimization*

- Replaced ~100 individual DB writes per claim with a single bulk API call — cut write operations by ~99%, reducing DB connection usage and latency across all services.
- Resolved recurring production timeouts: stale DB statistics caused full scans of 646K rows to return 179 results. Applied targeted indexes and maintenance — cut RDS CPU from ~80% to sub-30% and query execution time by 95%+.
- Optimized Report Summary API from 2+ minutes (or crashes) to a consistent 10–15s for 3-month ranges using streaming and parallelization — an 87%+ improvement.

LRD / DSD — Python Microservices *Healthcare document digitization*

- Built RESTful APIs using FastAPI/Uvicorn with Azure AI Form Recognizer and MongoDB for intelligent document processing workflows.
- Deployed production-grade Python microservices with graceful shutdown handling, CORS middleware, environment-based configuration, and Grafana telemetry.
- Worked on agentic AI workflows within the engineering team, improving developer productivity and reducing time spent on repetitive investigation and boilerplate tasks.

Frontend — Config-Driven Module System

React + Redux

- Identified that onboarding each new insurance client required duplicating ~29,000 lines of hand-written JSON config across 3 screens. Designed and built a V3 config-driven module system (69 TypeScript modules, transformer layer, registry pattern) reducing per-client config to ~500 lines — a 98% reduction.
- Preserved full backward compatibility with the existing React + Redux render pipeline. Delivered end-to-end across 20 phases with 780+ automated tests at 96%+ coverage.

Software Engineer — Mobil80 Solutions

Bangalore, India

HelloWrk · Posture.ai · StepSync

Jul 2023 – Aug 2024

- Led backend engineering for HelloWrk CRM — integrated SIP protocol for third-party call providers and built a full email client using AWS SES.
- Architected serverless infrastructure (Lambda, API Gateway, AppSync, SQS, DynamoDB, EventBridge) for multiple SaaS products.
- Built Posture.ai ML platform with OpenID integrations and automated health-concern detection algorithms.
- Led StepSync fitness app backend — resolved cross-platform fitness tracker compatibility and built centralised metrics tracking across watchOS devices.
- Led teams of 3–4 engineers; handled client discussions, stakeholder requirements, and end-to-end delivery.

Software Trainee — Bosch Global Software Technologies

Bangalore, India

Embedded Systems

Jan 2023 – May 2023

- Built a fully autonomous maze-solving robot using Arduino and C++; engineered a roll-over detection system with real-time tilt alerts.

PROJECTS

- **Pasteso — Markdown Code/Text Snippet Platform:** Full-stack web app (Node.js, React) with user management, public leaderboards, and snippet sharing. Reached **400K+ unique users** organically in one year. Exclusively on AWS — Lambda, API Gateway, DynamoDB, S3, CloudFront. Built and owned end-to-end including CI/CD pipelines from scratch.

CERTIFICATIONS & COURSES

- **Claude Code In Action:** Anthropic, Mar 2026 — verify.skilljar.com/c/zrici7when6o
- **Model Context Protocol: Introduction:** Anthropic, Mar 2026 — verify.skilljar.com/c/jd4r5ik4m2n6
- **Applied Software Engineering Fundamentals Specialization:** IBM via Coursera, Feb 2024
- **GATE 2026 — Data Science & AI:** Ranked **2323**. Topics covered: Statistics & Probability, Linear Algebra, Calculus, DSA, DBMS, Supervised & Unsupervised ML, Neural Networks, AI Search & Reasoning.